DATABASE MANAGEMENT SYSTEMS LAB

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LAB SHEET – 1 SOLUTION

Objective

To design Entity Relationship (ER) diagrams and convert entities and relationships into relational schemas. Further, to implement the schemas using SQL by applying appropriate constraints and performing DDL and DML operations.

PART A – COLLEGE DATABASE

Entities and Attributes

1. **STUDENT**

- USN (Primary Key)
- SName
- Address
- Phone
- Gender

2. **SEMSEC**

- SSID (Primary Key)
- Sem
- Sec

3. **CLASS**

- USN (Foreign Key)
- SSID (Foreign Key)

4. **SUBJECT**

- Subcode (Primary Key)
- Title
- Sem

- Credits

5. **IAMARKS**

- USN (Foreign Key)
- Subcode (Foreign Key)
- SSID (Foreign Key)
- Test1
- Test2
- Test3
- FinalIA

PART B – COMPANY DATABASE

Entities and Attributes

1. **EMPLOYEE**

- SSN (Primary Key)
- Name
- Address
- Sex
- Salary
- Superssn (Foreign Key)
- Dno (Foreign Key)

2. **DEPARTMENT**

- Dno (Primary Key)
- Dname
- Mgrssn
- Mgrstartdate

3. **DLOCATION**

- Dno (Foreign Key)
- Dloc

4. **PROJECT**

- PNo (Primary Key)
- PName
- PLocation
- DNo (Foreign Key)

5. **WORKS_ON**

- SSN (Foreign Key)
- PNo (Foreign Key)
- Hours

PART C – CUSTOMER SALE DATABASE

Entities

1. **CUSTOMER**

- Cust_id (Primary Key)
- cust_name

2. **ITEM**

- item_id (Primary Key)
- item_name
- price

3. **SALE**

- bill_no (Primary Key)
- bill_date
- cust_id (Foreign Key)
- item_id (Foreign Key)
- qty_sold

DDL QUESTIONS (SQL)

1. Create Database

```
CREATE DATABASE DBMS_AdityaWadhawan;
USE DBMS_AdityaWadhawan;
```

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2. Create Tables

STUDENT

```
CREATE TABLE STUDENT (
   USN VARCHAR(20) PRIMARY KEY,
```

```
    SName VARCHAR(50),
    Address VARCHAR(100),
    Phone VARCHAR(15),
    Gender CHAR(1)
);
```

SEMSEC

```
CREATE TABLE SEMSEC (
    SSID INT PRIMARY KEY,
    Sem INT,
    Sec CHAR(1)
);
```

CLASS

```
CREATE TABLE CLASS (
    USN VARCHAR(20),
    SSID INT,
    PRIMARY KEY (USN, SSID),
    FOREIGN KEY (USN) REFERENCES STUDENT(USN),
    FOREIGN KEY (SSID) REFERENCES SEMSEC(SSID)
);
```

SUBJECT

```
CREATE TABLE SUBJECT (
    Subcode VARCHAR(10) PRIMARY KEY,
    Title VARCHAR(50),
    Sem INT,
    Credits INT
);
```

IAMARKS

```
CREATE TABLE IAMARKS (
    USN VARCHAR(20),
    Subcode VARCHAR(10),
    SSID INT,
    Test1 INT,
    Test2 INT,
    Test3 INT,
    FinalIA INT,
    PRIMARY KEY (USN, Subcode, SSID),
    FOREIGN KEY (USN) REFERENCES STUDENT(USN),
    FOREIGN KEY (Subcode) REFERENCES SUBJECT(Subcode),
    FOREIGN KEY (SSID) REFERENCES SEMSEC(SSID)
);
```

Company Tables

EMPLOYEE

```
CREATE TABLE EMPLOYEE (
   SSN INT PRIMARY KEY,
   Name VARCHAR(50),
   Address VARCHAR(100),
   Sex CHAR(1),
   Salary INT NOT NULL,
   Superssn INT,
   Dno INT
);
```

DEPARTMENT

```
CREATE TABLE DEPARTMENT (
   Dno INT PRIMARY KEY,
   Dname VARCHAR(50),
   Mgrssn INT,
   Mgrstartdate DATE
);
```

DLOCATION

```
CREATE TABLE DLOCATION (
   Dno INT,
   Dloc VARCHAR(50),
   FOREIGN KEY (Dno) REFERENCES DEPARTMENT(Dno)
);
```

Customer Sale Tables

```
CREATE TABLE CUSTOMER (
   Cust_id INT PRIMARY KEY,
   cust_name VARCHAR(50)
);

CREATE TABLE ITEM (
item_id INT PRIMARY KEY,
item_name VARCHAR(50),
   price INT NOT NULL
);

CREATE TABLE SALE (
bill_no INT PRIMARY KEY,
   bill_date DATE,
cust_id INT,
item_id INT,
qty_sold INT,
   FOREIGN KEY (cust_id) REFERENCES CUSTOMER(Cust_id),
   FOREIGN KEY (item_id) REFERENCES ITEM(item_id)
);
```

DDL Operations

Add Email Column

ALTER TABLE STUDENT ADD Email VARCHAR(50);

Rename SName

ALTER TABLE STUDENT RENAME COLUMN SName TO StudentName;

Drop DLOCATION

DROP TABLE DLOCATION;

DML QUESTIONS

Insert Records

INSERT INTO CUSTOMER VALUES
(1,'Amit'),(2,'Riya'),(3,'Karan'),(4,'Neha'),(5,'Rahul'),
(6,'Pooja'),(7,'Ankit'),(8,'Simran'),(9,'Rohit'),(10,'Sonal');

INSERT INTO ITEM VALUES
(101,'Pen',50),(102,'Book',150),(103,'Bag',600),(104,'Bottle',250),
(105,'Shoes',1200),(106,'Notebook',100),(107,'Mouse',500),
(108,'Keyboard',800),(109,'Charger',300),(110,'Headphones',1500);

INSERT INTO SALE VALUES
(1,CURDATE(),5,103,2),
(2,CURDATE(),3,104,1),
(3,CURDATE(),5,105,1),
(4,CURDATE(),2,102,3),
(5,CURDATE(),1,101,5);

Queries

Bills of Current Date

SELECT bill_no, cust_name, item_name
FROM SALE
JOIN CUSTOMER ON SALE.cust_id = CUSTOMER.Cust_id
JOIN ITEM ON SALE.item_id = ITEM.item_id
WHERE bill_date = CURDATE();

Total Bill Amount

```
SELECT bill_no, qty_sold * price AS TotalAmount
FROM SALE
JOIN ITEM ON SALE.item_id = ITEM.item_id;
```

Customers Purchasing Items Above ₹200

```
SELECT DISTINCT cust_name
FROM CUSTOMER
JOIN SALE ON CUSTOMER.Cust_id = SALE.cust_id
JOIN ITEM ON SALE.item_id = ITEM.item_id
WHERE price > 200;
```

Count Products per Customer

```
SELECT cust_name, COUNT(item_id) AS TotalProducts
FROM CUSTOMER
JOIN SALE ON CUSTOMER.Cust_id = SALE.cust_id
GROUP BY cust_name;
```

Products Purchased by Cust_id = 5

```
SELECT item_name
FROM SALE
JOIN ITEM ON SALE.item_id = ITEM.item_id
WHERE cust_id = 5;
```

Views

Bill Details View

```
CREATE VIEW Bill_Details AS
SELECT bill_no, bill_date, cust_id, item_id, price, qty_sold,
(price * qty_sold) AS Amount
FROM SALE
JOIN ITEM ON SALE.item_id = ITEM.item_id;
```

Daily Sales (Last One Week)

```
CREATE VIEW Weekly_Sales AS
SELECT bill_date, SUM(price * qty_sold) AS TotalSales
FROM SALE
JOIN ITEM ON SALE.item_id = ITEM.item_id
WHERE bill_date >= CURDATE() - INTERVAL 7 DAY
GROUP BY bill_date;
```

Conclusion

Thus, ER diagrams were designed, converted into relational schemas, and implemented successfully using SQL. All DDL and DML operations were executed with proper constraints, ensuring data integrity and normalization.